

CSE 4849/CSE 4885  
Human Computer Interaction (HCI)  
Lecture 4: Interaction Design

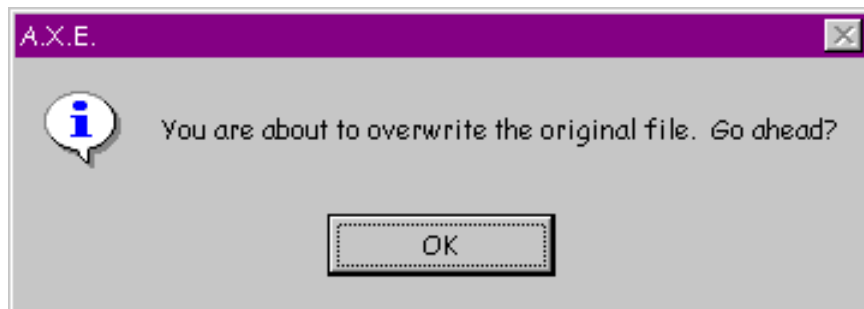
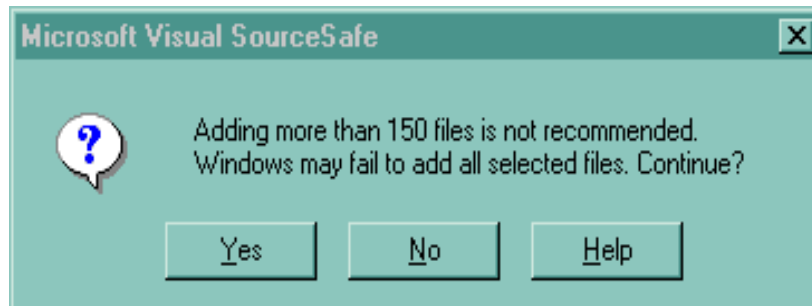
Summer 2025

Dr. Hasan Mahmud | [hasan@iut-dhaka.edu](mailto:hasan@iut-dhaka.edu)

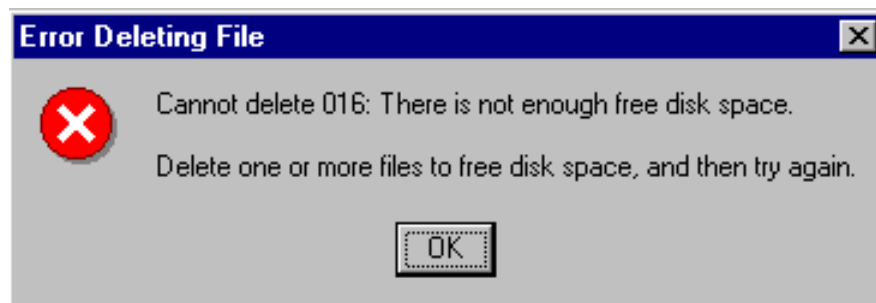
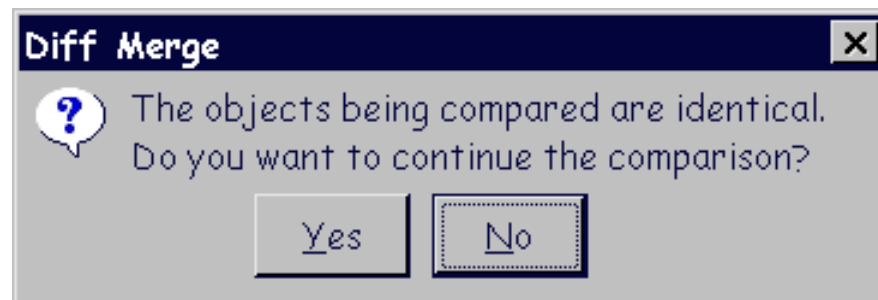
# Interaction Design

1. Pathology of Design
2. What is design?
3. What to design?
4. What is Interaction Design?
5. What is involved in the process of interaction design?
6. Goals of Interaction Design
  - a. Usability goal
  - b. User experience goal
7. Interaction Design and HCI

# Insane Dialog Boxes



# Insane Dialog Boxes



# Bad designs

- Elevator controls and labels on the bottom row all look the same, so it is easy to push a label by mistake instead of a control button



- People do not make same mistake for the labels and buttons on the top row. Why not?

From: [www.baddesigns.com](http://www.baddesigns.com)

# Why is this vending machine so bad?



From: [www.baddesigns.com](http://www.baddesigns.com)

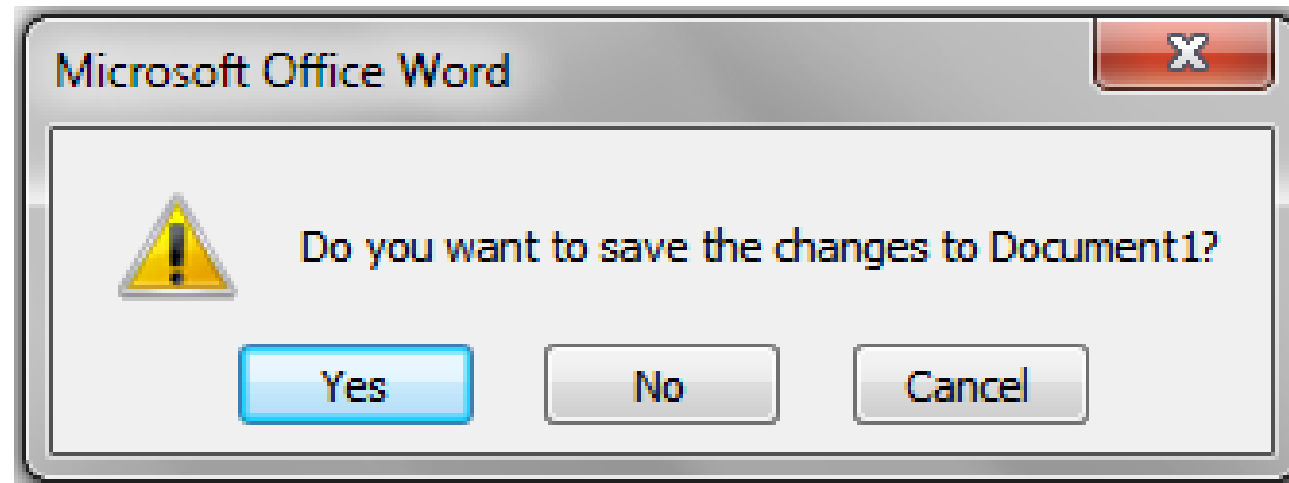
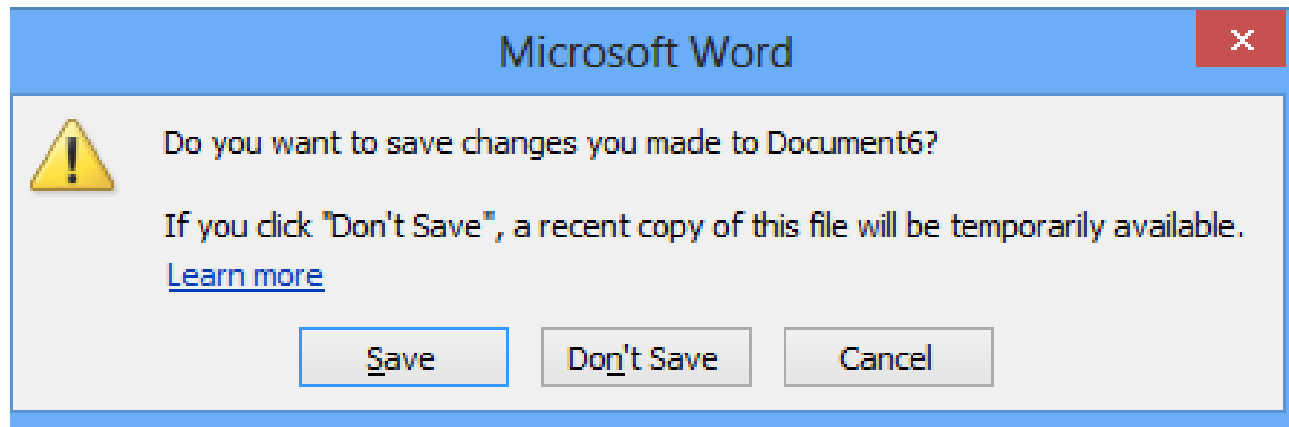
- Need to **push button first** to activate reader
- Normally **insert bill** first before making selection
- Contravenes well known convention

# Good and bad design



- What is wrong with the Apex remote?
- Why is the TiVo remote so much better designed?
  - Peanut shaped to fit in hand
  - Logical layout and colour-coded, distinctive buttons
  - Easy to locate buttons

# Good and bad design





# What is design?

*Achieving goals within constraints*

- Goals
  - What is the purpose of the design intended?
  - who is it for?
  - why do they want it ?
- Constraints
  - materials, platforms, standards, cost, time, health & safety etc.
- Trade-offs
  - Choosing between goals and constraints

# A Note on the Term “Design”

- Industrial Design
  - A branch of Architecture dealing with the design of artifact
- Graphical Design (or Visual Design)
  - A set of techniques for making effective visual presentations
  - Often requires a lot of “artistic” taste
- Interface Design
  - A set of principles (or even guidelines) to design computer graphical interfaces (GUI)

- User-Centred Design (UCD)
  - A term introduced by Norman in the '80
  - Refers to the need of considering the user in the life-cycle of software
- Interaction Design
  - A trend in Industrial Design that emphasis the role of the user
  - In IT, it often refers to UCD after the GUI era
- Participatory Design
  - A Swedish technique to implement extreme UCD

# What to design?

- Need to take into account:
  - Who the users are
  - What activities are being carried out
  - Where the interaction is taking place
- The appropriateness of different kinds of interfaces and arrangements of input and output devices depends on what ***kinds of activities*** need to be supported.
- The range of activities that can be supported is diverse.
- There are also many ways of designing the way users can interact with a system.
- Need to optimize the interactions users have with a product
  - So that they match the users' activities and needs

# Understanding the users..

- Need to take into account what people are good and bad at
- Consider what might help people in the way they currently do things
- Think through what might provide quality user experiences
- Listen to what people want and get them involved
- Use tried and tested user-centred methods

# Activity

- How does making a call differ when using a:
  - Cell phone
  - Public phone box?
- Consider the kinds of user, type of activity and context of use



# Anna, IKEA online sales agent

- Designed to be different for UK and US customers
- What are the differences and which is which?
- What should Anna's appearance be like for other countries, like India, South Africa, or China?



# golden rule of design

Understand your materials



# for Human–Computer Interaction

## Understand your materials

- Understand computers
  - Limitations, capacities, tools, platforms
- Understand people
  - Psychological, social aspects
  - Human error
- And their interaction ...

# What is interaction design?

- ***Designing interactive products to support people in their everyday and working lives.***
  - it is about creating user experiences that enhance and extend the way people work, communicate and interact
    - Sharp, Rogers and Preece (2007)
- ***The design of spaces for human communication and interaction*** – Winograd (1997)
  - finding way of supporting people
  - Architects vs. civil engineers
- ***The why as well as the how of our daily interactions using computers*** – Thackara (2001)

# What is involved in the process of interaction design

- Identifying needs and establishing requirements for the user experience
- Developing alternative designs to meet these
- Building interactive prototypes that can be communicated and assessed
- Evaluating what is being built throughout the process and the user experience it offers

# Core characteristics of interaction design

- Users should be involved through the development of the project
- Specific usability and user experience goals need to be identified, clearly documented and agreed at the beginning of the project
- Iteration is needed through the core activities

# Goals of interaction design

- Usability goals
  - Develop usable products
    - Usability means easy to learn, effective to use and provide an enjoyable experience
  - Goals are concerned with meeting specific usability criteria (e.g., efficiency)
- User experience goals
  - User experience goals are largely concerned with explicating the quality of the user experience (e.g., to be aesthetically pleasing).

# Usability goals

- Effective to use
  - Refers to how good a system is at doing what it is supposed to do.
- Efficient to use
  - Refers to the way a system supports users in carrying out their tasks. Let the user carry out common tasks through a minimal number of steps. E.g. one-click option patented by Amazon.com
- Safe to use
  - Involves protecting the user from dangerous conditions and undesirable situations
    - preventing the user from making serious errors by reducing the risk of wrong keys/ buttons being mistakenly activated. An example is not placing the quit or delete-file command right next to the save command on a menu
    - providing users with various means of recovery should they make errors.
    - Other safety mechanisms include undo facilities and confirmatory dialog boxes that give users another chance to consider their intentions.

# Usability goals

- Have good utility
  - Refers to the extent to which the system provides the right kind of functionality so that users can do what they need or want to do.
  - An example of a system with high utility is an accounting software package providing a powerful computational tool that accountants can use to work out tax returns.
- Learnability
  - Refers to how easy a system is to learn to use.
  - A key concern is determining how much time users are prepared to spend learning a system.
  - Flash tutorial vs. Interactive TV, email
- Memorability
  - Refers to how easy a system is to remember how to use, once learned.

# How to Measure Usability?

- We want to achieve these goals, but how do we know?
- Develop measurable criteria based on previous goals. Examples:
  - Time to learn
  - Speed of performance
  - Rate of errors over by users
  - Retention over time
  - Subjective satisfaction



# User experience goals

- satisfying
- enjoyable
- engaging
- pleasurable
- exciting
- entertaining
- helpful
- motivating
- emotionally fulfilling
- aesthetically pleasing
- supportive of creativity
- supportive of creativity
- rewarding
- fun
- provocative
- surprising
- enhancing sociability
- challenging
- boring
- frustrating
- annoying
- cutsey

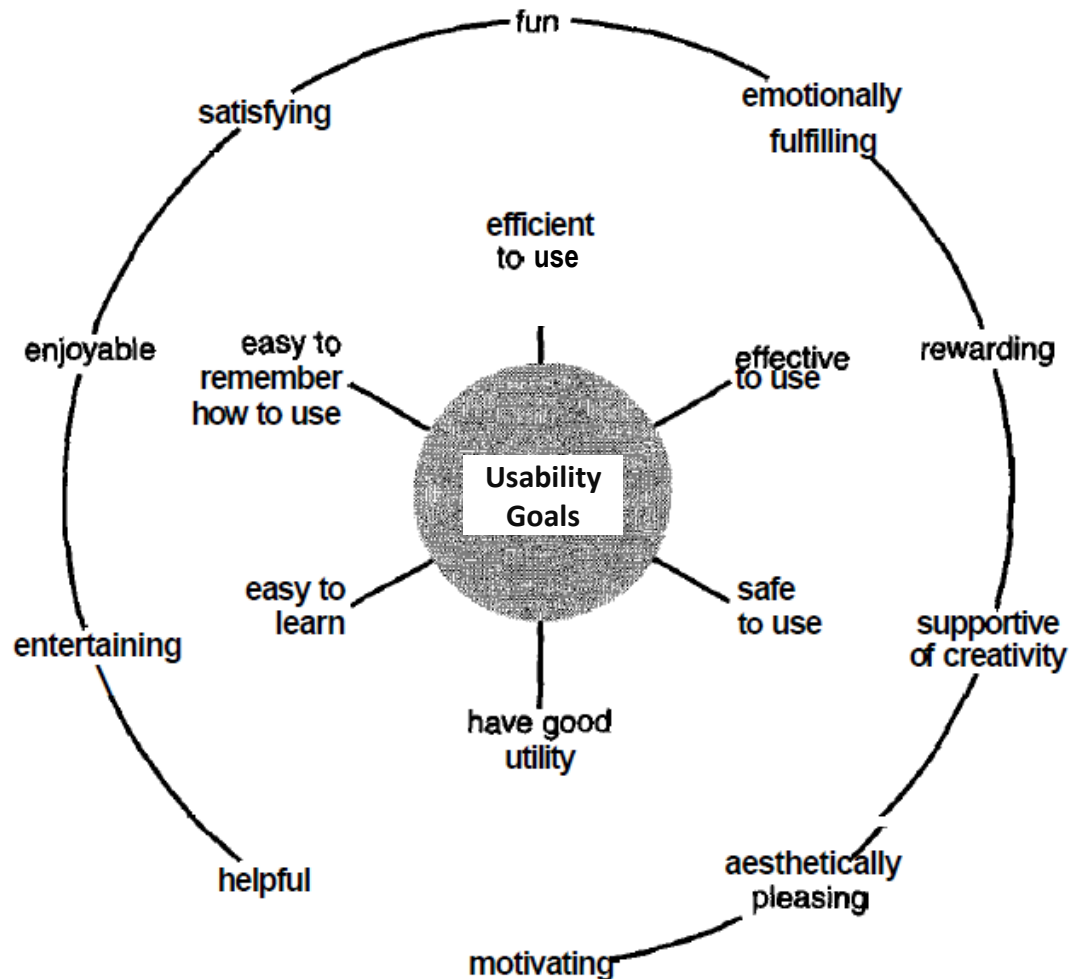
# Usability and user experience goals

- Selecting terms to convey a person's feelings, emotions, etc., can help designers understand the multifaceted nature of the user experience
- How do usability goals differ from user experience goals?
- Are there trade-offs between the two kinds of goals?
  - e.g. can a product be both fun and safe?
- How easy is it to measure usability versus user experience goals?

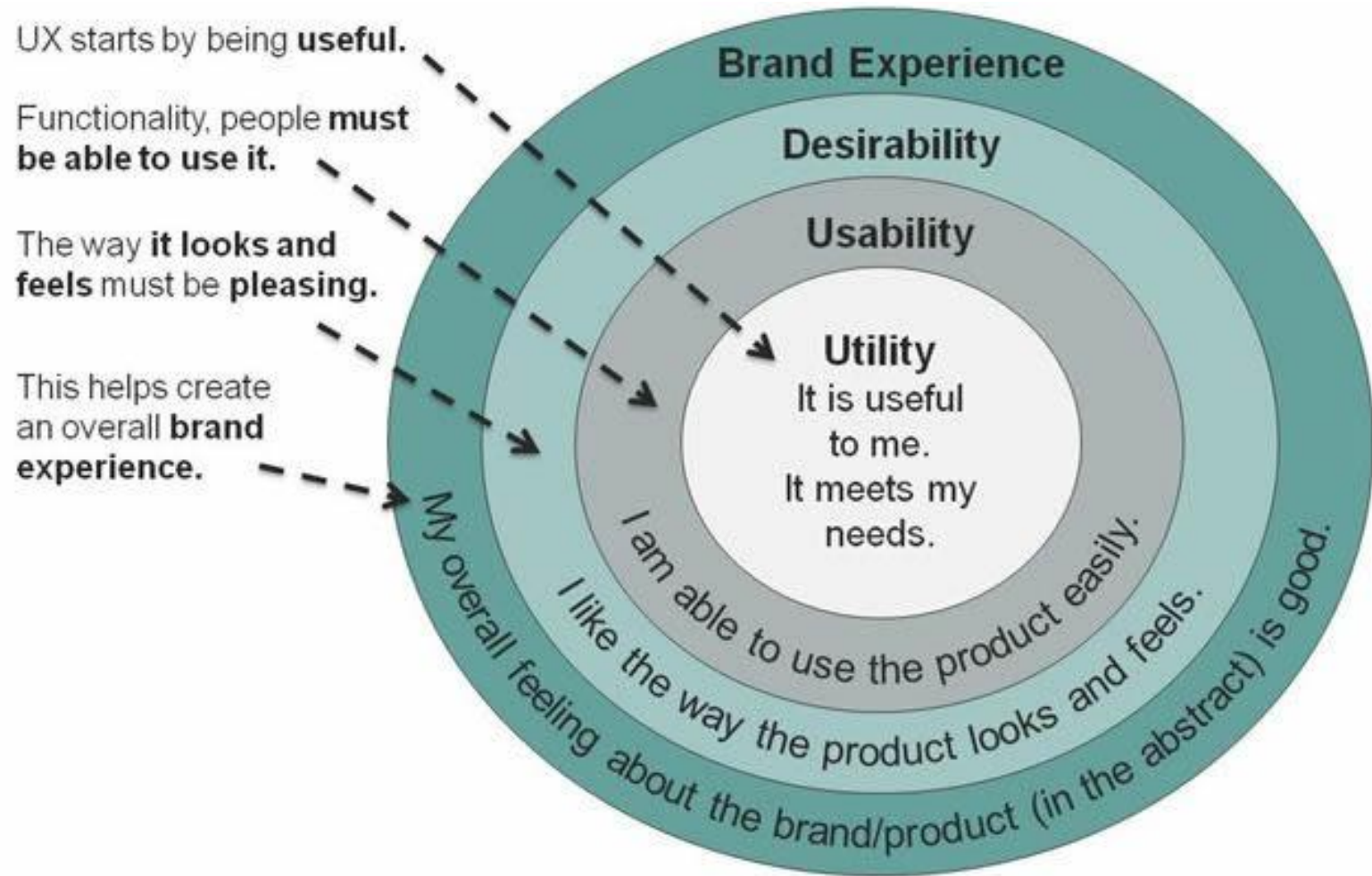
# The User Experience

- How a product behaves and is used by people in the real world
  - the way people feel about it and their pleasure and satisfaction when using it, looking at it, holding it, and opening or closing it
  - “every product that is used by someone has a user experience: newspapers, ketchup bottles, reclining armchairs, cardigan sweaters.” (Garrett, 2003)
- Cannot design a user experience, only design *for* a user experience

# Usability and user experience goals



# Usability and user experience - relationship



Source: User Experience 2008, nnGroup Conference Amsterdam  
Retrieved from: <http://neospot.se/usability-vs-user-experience/>

# Why was the iPod user experience such a success?



# Design principles

- Generalizable abstractions for thinking about different aspects of design
- The do's and don'ts of interaction design
  - But at a high level. (Compare to guidelines later.)
- What to provide and what not to provide at the interface
- Derived from a mix of theory-based knowledge, experience and common-sense

# Higher-level Principles

- Now, we will talk about the following:
  - Visibility: Can is see it?
  - Feedback: What is it doing now?
  - Constraints: Why can't I do that?
  - Mapping: Where am I and where can I go?
  - Consistency: I think I have seen this before?
  - Affordances: How do I use it?
- Ideas well-known (e.g. from *Norman's Design of Everyday Things*)



# Activity (Mini-Homework)

- Do this on your own or in pairs
- Go find two examples of problems in the GUIs of software apps or the UI of an interactive device (not Web pages)
  - Problem must illustrate a violation of one of the design principles or usability guidelines
  - Describe the problems in these terms

# Visibility



- This is a control panel for an elevator.
- How does it work?
- Push a button for the floor you want?
- Nothing happens. Push any other button? Still nothing. What do you need to do?

It is not visible as to what to do!

From:  
[www.baddesigns.com](http://www.baddesigns.com)

# Visibility



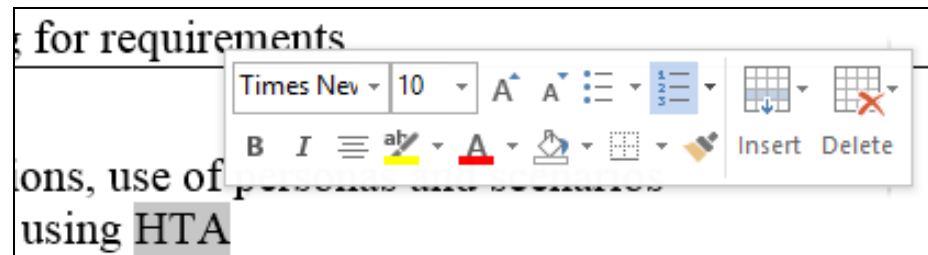
...you need to insert your room card in the slot by the buttons to get the elevator to work!

How would you make this action more **visible**?

- make the card reader more obvious
- provide an auditory message, that says what to do (which language?)
- provide a big label next to the card reader that flashes when someone enters
- make relevant parts visible
- make what has to be done obvious

# Visibility

- Hiding certain functions can be advantageous in interface design
- Certain functions are kept invisible until needed; also contained within a group of similar types

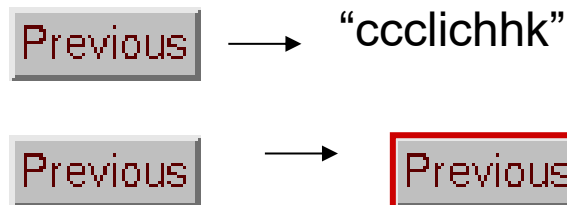


- Google search makes it clear where to enter text



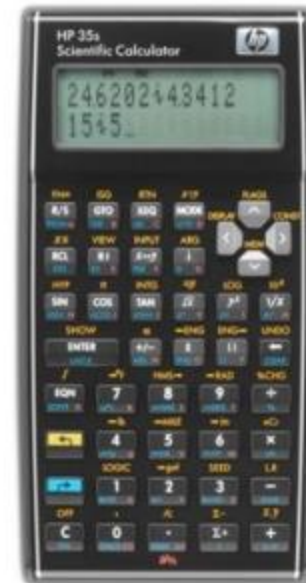
# Feedback

- What is it doing now? what action has been performed?
- Sending information back to the user about what has been done
- Includes sound, highlighting, animation and combinations of these
  - e.g. when screen button clicked on provides sound or red highlight feedback:



# Feedback

- Other examples



# Constraints

- Restricting the possible actions that can be performed
- Helps prevent user from selecting incorrect options, reduce the chance of doing error
- Can also work to focus user's attention to needed task
- Three main types (Norman, 1999)
  - physical
  - cultural
  - logical

# Physical constraints

- Refer to the way physical objects restrict the movement of things
  - E.g. only one way you can insert a key into a lock
- How many ways can you insert a CD or DVD disk into a computer?
- How physically constraining is this action?
- How does it differ from the insertion of a floppy disk into a computer?



# Logical constraints

- Exploits people's everyday common sense reasoning about the way the world works
- An example is the logical relationship between physical layout of a device and the way it works
  - See next slide for an illustration

# Logical or ambiguous design?



- Where do you plug the mouse?
- Where do you plug the keyboard?
- top or bottom connector?
- Do the color coded icons help?

From: [www.baddesigns.com](http://www.baddesigns.com)

# More Logically Constrained



← Provides direct adjacent mapping between icon and connector



← Provides color coding to associate the connectors with the labels

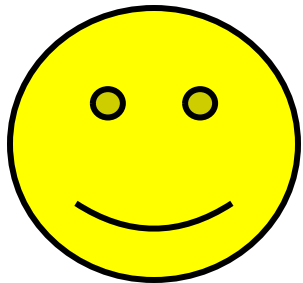
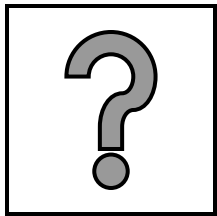
From: [www.baddesigns.com](http://www.baddesigns.com)

# Cultural constraints

- Learned arbitrary conventions like red triangles for warning
- Can be universal or culturally specific
  - For SW we've accepted certain conventions, e.g. we know what to do with an icon
- Be concerned of cross-cultural conventions and other ambiguities!
  - Does an "X" mean "selected" or "not selected"
  - Is a check-mark better?

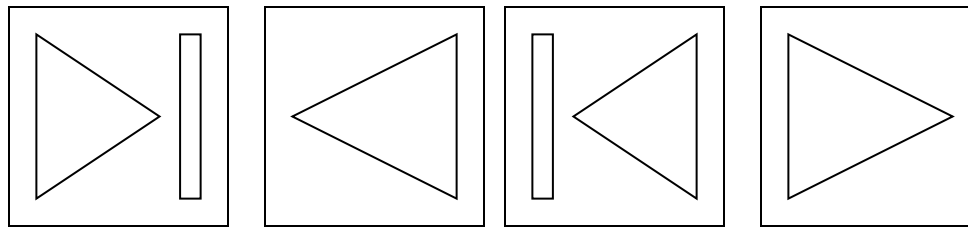


# Which are universal and which are culturally-specific?



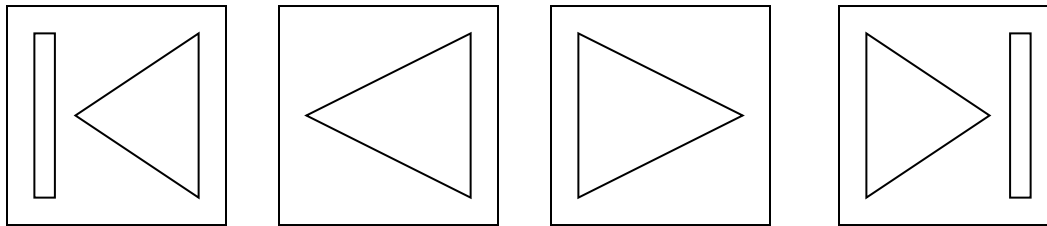
# Mapping

- Relationship between controls and their movements and the results in the world
- Why is this a poor mapping of control buttons?



# Mapping

- Why is this a better mapping?



- The control buttons are mapped better onto the sequence of actions of fast rewind, rewind, play and fast forward
  - Is this a logical mapping (in most people's minds)?
  - Is there a mapping that makes more sense?

# Mappings in the Kitchen

- Which controls go with which rings (burners)?



A



B



C



D



# Why is this a better design?



# Consistency

- Design interfaces to have similar operations and use similar elements for similar tasks
- For example:
  - always use ctrl key plus first initial of the command for an operation – ctrl+C, ctrl+S, ctrl+O
- Main benefit is consistent interfaces are easier to learn and use
- But... didn't a wise man say this?  
Consistency is the hobgoblin of little minds.
  - A foolish consistency is the hobgoblin of little minds (adored by little statesmen and philosophers and divines)

# When consistency breaks down

- What happens if there is more than one command starting with the same letter?
  - e.g. save, spelling, select, style
- Have to find other initials or combinations of keys, thereby breaking the consistency rule
  - E.g. ctrl+S, ctrl+Sp, ctrl+shift+L
- Is this desirable? Does it defeat the purpose?
  - It may increase learning burden on user
  - It may make them more prone to errors
  - But it may still benefit frequent or experienced users

# Internal and external consistency

- **Internal consistency** refers to designing operations to behave the same within an application
  - Difficult to achieve with complex interfaces
- **External consistency** refers to designing operations, interfaces, etc., to be the same across applications and devices
  - Very rarely the case, based on different designer's preference
  - Most successful in product families (e.g MS Office)
  - Op. Sys. vendors may define style guidelines

# Keypad numbers layout

- A case of external inconsistency

(a) phones, remote controls

|   |   |   |
|---|---|---|
| 1 | 2 | 3 |
| 4 | 5 | 6 |
| 7 | 8 | 9 |
|   | 0 |   |

(b) calculators, computer keypads

|   |   |   |
|---|---|---|
| 7 | 8 | 9 |
| 4 | 5 | 6 |
| 1 | 2 | 3 |
| 0 |   |   |

# Affordances: to give a clue

- Refers to an attribute of an object that allows people to know how to use it
  - E.g. a mouse button invites pushing, a door handle affords pulling
- Norman (1988) used the term to discuss the design of everyday objects
- Since then has been popularized in interaction design to discuss how to design interface objects
  - E.g. scrollbars to afford moving up and down, icons to afford clicking on

# What does 'affordance' have to offer interaction design?

- Notion of affordance is often over-used!  
<http://www.jnd.org/dn.mss/affordances-and.html>
- Interfaces are virtual and do not have affordances like physical objects
- Norman argues it does not make sense to talk about interfaces in terms of 'real' affordances
- Instead interfaces are better conceptualised as 'perceived' affordances
  - Learned conventions of arbitrary mappings between action and effect at the interface
  - Some mappings are better than others

# Activity 1.3

## – Physical affordances:

How do the following physical objects afford? Are they obvious?





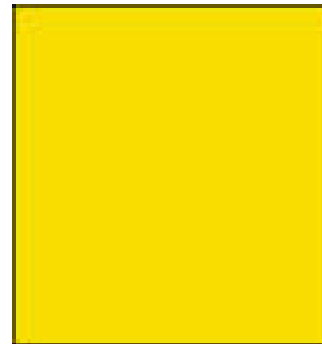
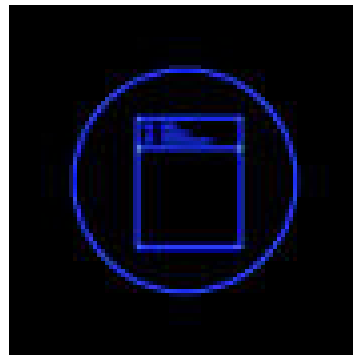
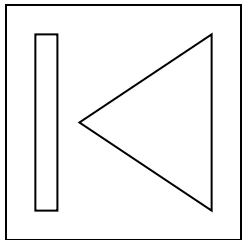
# Activity 1.4

## – Virtual affordances

How do the following screen objects afford?

What if you were a novice user?

Would you know what to do with them?



Up Arrow →

Elevator Button

Down Arrow →



# Activity 1.5: Physical and Perceived Affordances

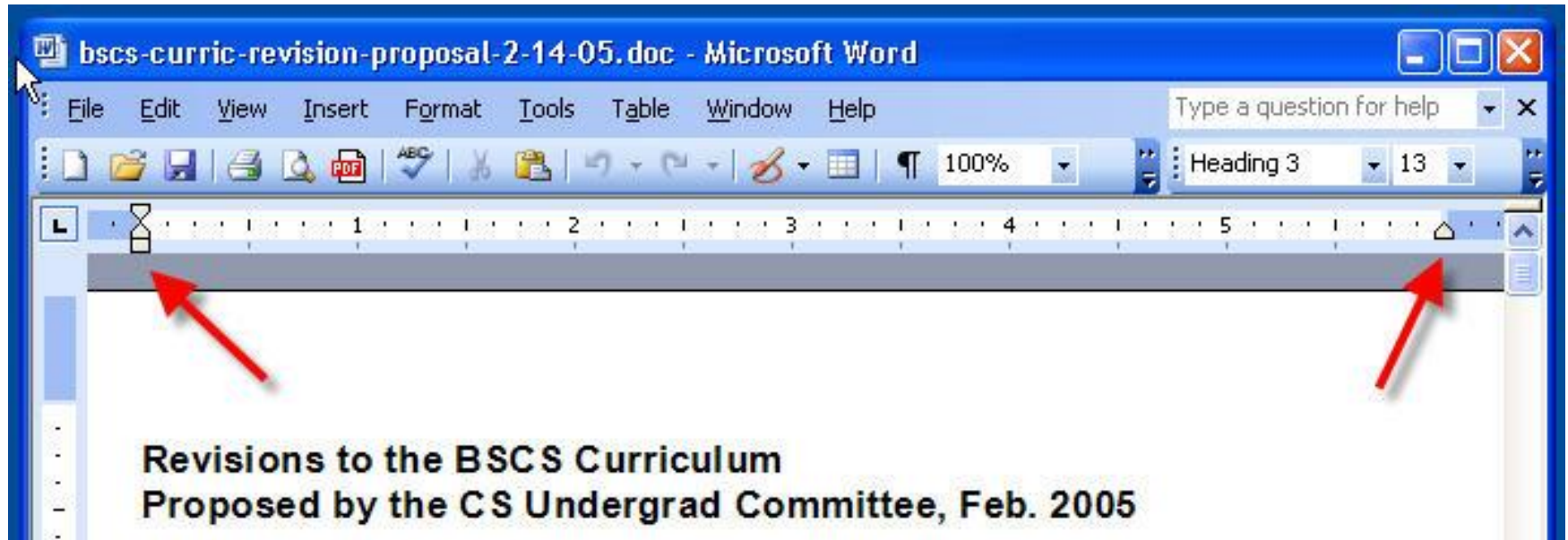
- Take a cell phone, digital camera, or PDA
  - Have laptop? Open a fancy SW app: Outlook, Eclipse, etc?
- In a small group
  - Identify any physical affordances the device has
  - Identify any perceived or visual affordances the software user interface has
- Write these down, be prepared to share or turn in

# A Good Example

- Kodak DC-290 digital camera



# Adjusting Tabs in MS Word



- What's the idea here? Problems?
- What principle(s)?
  - Affordance
  - Metaphor

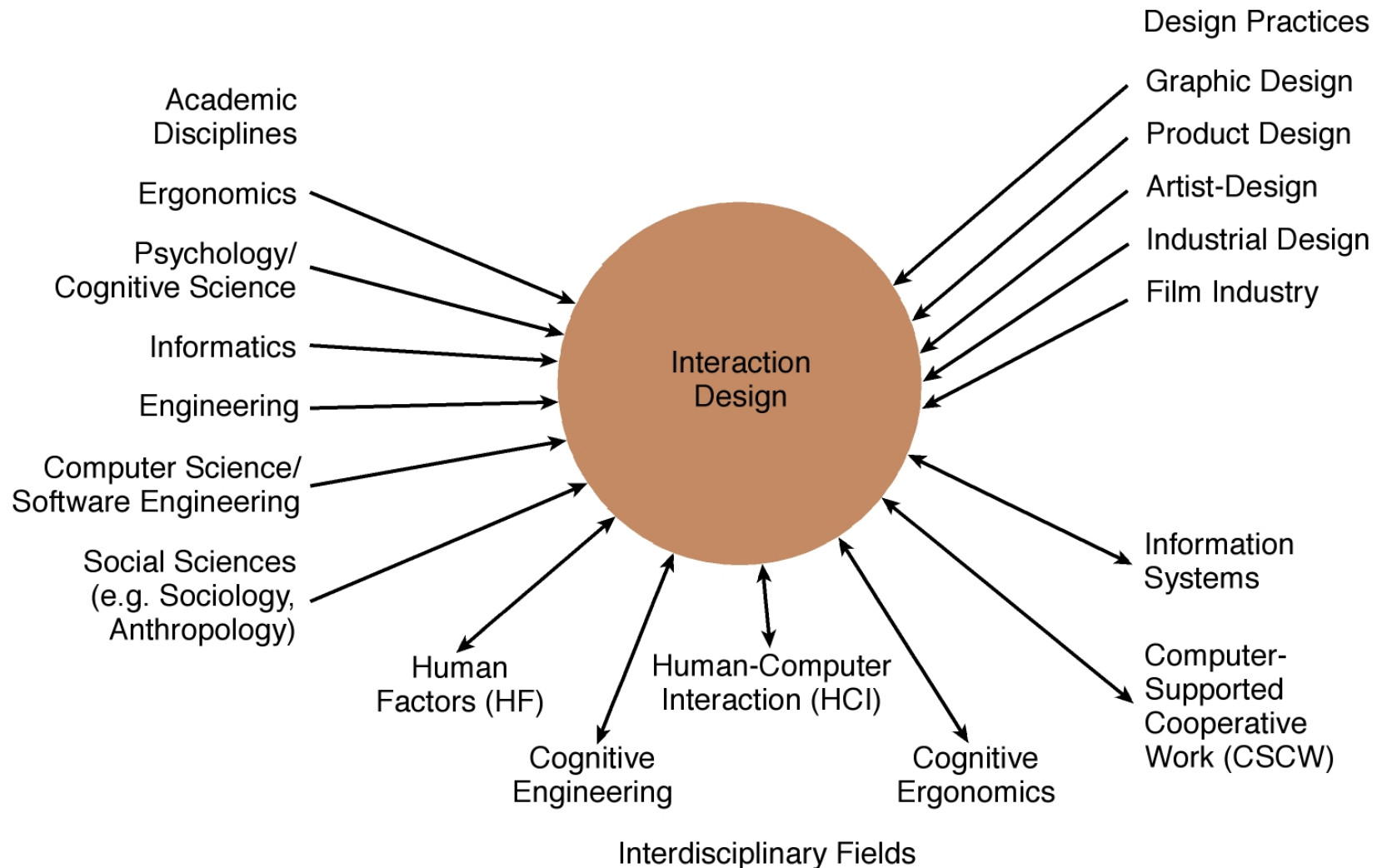
# “Usability principles” (Nielsen 2001)

- Visibility of system status
- Match between system and the real world
- User control and freedom
- Consistency and standards
- Help users recognize, diagnose and recover from errors
- Error prevention
- Recognition rather than recall
- Flexibility and efficiency of use
- Aesthetic and minimalist design
- Help and documentation

# Web Links

- What are the conventions that help you recognize a link?
- Would you argue this is an affordance?
  - A perceived affordance – convention of a mapping between action and effect
  - Does it “afford” clicking on it?
- Examples of problems with this?
  - Let’s look at: <http://www.cs.virginia.edu>

# HCI and interaction design



# Relationship between ID, HCI and other fields

- Academic disciplines contributing to ID:
  - Psychology
  - Social Sciences
  - Computing Sciences
  - Engineering
  - Ergonomics
  - Informatics



# Relationship between ID, HCI and other fields

- Design practices contributing to ID:
  - Graphic design
  - Product design
  - Artist-design
  - Industrial design
  - Film industry

# Relationship between ID, HCI and other fields

- Interdisciplinary fields that 'do' interaction design:
  - HCI
  - Human Factors
  - Cognitive Engineering
  - Cognitive Ergonomics
  - Computer Supported Co-operative Work
  - Information Systems

# Working in multidisciplinary teams

- Many people from different backgrounds involved
- Different perspectives and ways of seeing and talking about things
- Benefits
  - more ideas and designs generated
- Disadvantages
  - difficult to communicate and progress forward the designs being create



# Interaction design in business

- Increasing number of ID consultancies, examples of well known ones include:
  - **Nielsen Norman Group**: “help companies enter the age of the consumer, designing human-centered products and services”
  - **Cooper**: “From research and product to goal-related design”
  - **Swim**: “provides a wide range of design services, in each case targeted to address the product development needs at hand”
  - **IDEO**: “creates products, services and environments for companies pioneering new ways to provide value to their customers”



# What do professionals do in the ID business?

- **interaction designers** - people involved in the design of all the interactive aspects of a product
- **usability engineers** - people who focus on evaluating products, using usability methods and principles
- **web designers** - people who develop and create the visual design of websites, such as layouts
- **information architects** - people who come up with ideas of how to plan and structure interactive products
- **user experience designers (UX)** - people who do all the above but who may also carry out field studies to inform the design of products

# Readings

- Chapter 5: Interaction Design basics – Alan dix
- Chapter 1: What is interaction design - Sharp